

## Lecture 22: Line Integrals

### Background & Motivation

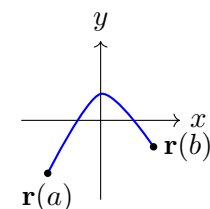
With vector fields defined, we now integrate them along curves. The line integral  $\int_C \mathbf{F} \cdot d\mathbf{r}$  computes the work done by a force along a path. For conservative fields, the Fundamental Theorem of Line Integrals gives us a shortcut: the integral depends only on the endpoints, not the path.

### 1. Contour Types

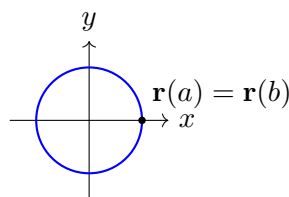
A **contour** (or **curve**)  $C$  is a path in  $\mathbb{R}^2$  or  $\mathbb{R}^3$  given by a parameterization  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ .

#### Definition 1: Types of Contours

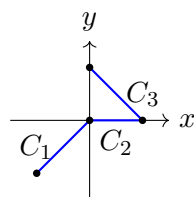
- **Open** – endpoints  $\mathbf{r}(a) \neq \mathbf{r}(b)$
- **Closed** –  $\mathbf{r}(a) = \mathbf{r}(b)$  (loop); use notation  $\oint_C$
- **Simple** – does not cross itself
- **Piecewise smooth** – finitely many smooth pieces joined end-to-end



Open Contour



Closed Contour



Piecewise Contour

### Common Parameterizations

- Line segment from  $A$  to  $B$  –  $\mathbf{r}(t) = (1-t)A + tB$ ,  $t \in [0, 1]$
- Circle  $x^2 + y^2 = R^2$  –  $\mathbf{r}(t) = \langle R \cos t, R \sin t \rangle$ ,  $t \in [0, 2\pi]$
- Graph  $y = f(x)$ ,  $a \leq x \leq b$  –  $\mathbf{r}(t) = \langle t, f(t) \rangle$ ,  $t \in [a, b]$
- Helix –  $\mathbf{r}(t) = \langle a \cos t, a \sin t, bt \rangle$

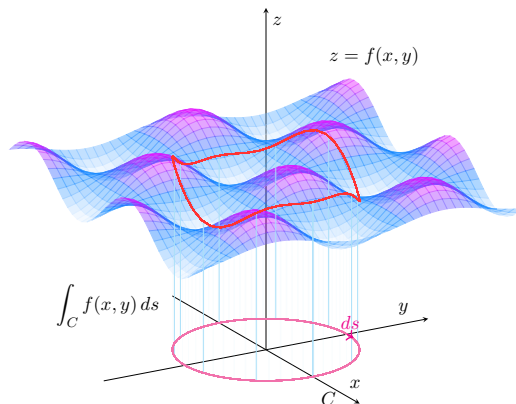
### 2. Scalar Line Integrals

## Definition 2: Scalar Line Integral

For a scalar function  $f$  along a curve  $C$  parameterized by  $\mathbf{r}(t)$ :

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt$$

- $ds = \|\mathbf{r}'(t)\| \, dt$  is the **arc length element**
- **Interpretation:** area of the “curtain” under  $f$  along  $C$
- Direction does not matter:  $\int_C f \, ds = \int_{-C} f \, ds$



## Example 1: Scalar Line Integral Along a Line Segment

Evaluate  $\int_C (x + 2y) \, ds$  where  $C$  is the segment from  $(0, 0)$  to  $(3, 4)$ .

1. **Parameterize:**  $\mathbf{r}(t) = \langle 3t, 4t \rangle$ ,  $t \in [0, 1]$
2. **Speed:**  $\mathbf{r}'(t) = \langle 3, 4 \rangle$ ,  $\|\mathbf{r}'(t)\| = \sqrt{9 + 16} = 5$
3. **Substitute:**  $f(\mathbf{r}(t)) = 3t + 2(4t) = 11t$
4. **Integrate:**  $\int_0^1 11t \cdot 5 \, dt = 55 \int_0^1 t \, dt = 55 \cdot \frac{1}{2} = \boxed{\frac{55}{2}}$

## 3. Vector Field Line Integrals

### Definition 3: Line Integral of a Vector Field

For a vector field  $\mathbf{F}$  along a curve  $C$ :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$

- **Interpretation:** work done by the force  $\mathbf{F}$  along path  $C$
- Reversing direction **negates** the integral:  $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$
- Equivalent notation:  $\int_C P \, dx + Q \, dy + R \, dz$

### Example 2: Work Around a Circle

Compute  $\int_C \mathbf{F} \cdot d\mathbf{r}$  for  $\mathbf{F} = \langle -y, x \rangle$  along the circle  $x^2 + y^2 = 4$  (CCW).

1. **Parameterize:**  $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$ ,  $\mathbf{r}' = \langle -2 \sin t, 2 \cos t \rangle$ ,  $t \in [0, 2\pi]$
2. **Evaluate  $\mathbf{F}$ :**  $\mathbf{F}(\mathbf{r}(t)) = \langle -2 \sin t, 2 \cos t \rangle$
3. **Dot product:**  $\mathbf{F} \cdot \mathbf{r}' = (-2 \sin t)(-2 \sin t) + (2 \cos t)(2 \cos t) = 4 \sin^2 t + 4 \cos^2 t = 4$
4. **Integrate:**  $\int_0^{2\pi} 4 \, dt = \boxed{8\pi}$

## 4. Fundamental Theorem of Line Integrals (FTOLI)

### Theorem 1: Fundamental Theorem of Line Integrals

If  $\mathbf{F} = \nabla f$  (conservative) and  $C$  goes from  $A$  to  $B$ :

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

#### Consequences:

- The integral depends only on **the endpoints** – **path independence**
- For any closed curve:  $\oint_C \nabla f \cdot d\mathbf{r} = 0$
- If  $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$  for some closed  $C$ , then  $\mathbf{F}$  is **not** conservative

### Example 3: Applying FTOLI

Evaluate  $\int_C \langle ye^{xy}, xe^{xy} \rangle \cdot d\mathbf{r}$  from  $(0,0)$  to  $(1,2)$ .

1. **Recognize:** From Worksheet 21,  $\mathbf{F} = \nabla f$  with  $f(x, y) = e^{xy}$
2. **Apply FTOLI:**  $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = e^{(1)(2)} - e^0 = \boxed{e^2 - 1}$

No parameterization needed — any path from  $(0,0)$  to  $(1,2)$  gives the same answer.

### Example 4: Piecewise 3D Line Integral

Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$  for  $\mathbf{F} = \langle y^2, z^2, x^2 \rangle$  along the path  $(0,0,0) \rightarrow (1,0,0) \rightarrow (1,1,0) \rightarrow (1,1,1)$ .

**Segment 1:**  $(0,0,0) \rightarrow (1,0,0)$ :  $\mathbf{r} = \langle t, 0, 0 \rangle$ ,  $\mathbf{r}' = \langle 1, 0, 0 \rangle$

$\mathbf{F} = \langle 0, 0, t^2 \rangle$ ,  $\mathbf{F} \cdot \mathbf{r}' = 0$ . Integral = 0.

**Segment 2:**  $(1,0,0) \rightarrow (1,1,0)$ :  $\mathbf{r} = \langle 1, t, 0 \rangle$ ,  $\mathbf{r}' = \langle 0, 1, 0 \rangle$

$\mathbf{F} = \langle t^2, 0, 1 \rangle$ ,  $\mathbf{F} \cdot \mathbf{r}' = 0$ . Integral = 0.

**Segment 3:**  $(1,1,0) \rightarrow (1,1,1)$ :  $\mathbf{r} = \langle 1, 1, t \rangle$ ,  $\mathbf{r}' = \langle 0, 0, 1 \rangle$

$\mathbf{F} = \langle 1, t^2, 1 \rangle$ ,  $\mathbf{F} \cdot \mathbf{r}' = 1$ . Integral =  $\int_0^1 1 \, dt = 1$ .

**Total:**  $0 + 0 + 1 = \boxed{1}$

## Practice Problems

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**Problem 1.** Evaluate  $\int_C xy \, ds$  where  $C$  is the line segment from  $(0,0)$  to  $(1,1)$ .

Answer:  $\frac{\sqrt{2}}{3}$

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**Problem 2.** Evaluate  $\int_C \langle y^2, 2xy \rangle \cdot d\mathbf{r}$  from  $(0,0)$  to  $(1,1)$  along  $y = x^2$ .

Answer: 1

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**Problem 3.** Evaluate  $\oint_C \langle x^2, y^2 \rangle \cdot d\mathbf{r}$  where  $C$  is the unit circle. (*Hint: is the field conservative?*)

Answer: 0

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**Problem 4.** Evaluate  $\int_C \langle 2xz + y, x + z^2, x^2 + 2yz \rangle \cdot d\mathbf{r}$  from  $(0,0,0)$  to  $(1,2,3)$  using FTOLI.

(*Hint: find the potential function first.*)

Answer:  $f = x^2z + xy + yz^2$ ;  $f(1,2,3) - f(0,0,0) = 3 + 2 + 18 = 23$

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**Problem 5.** Evaluate  $\int_C (x^2 + y^2) ds$  where  $C$  is the helix  $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ ,  $0 \leq t \leq 2\pi$ .

Answer:  $2\sqrt{2}\pi$